

A SHORT TUTORIAL ON GRADIENT PATTERN ANALYSIS FOR VISUAL COMPUTING

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ABSTRACT SUBMITTED TO ICCV23-IEEE COMPUTER SOCIETY, PARIS-2023

This TUTORIAL is structured in theory and practice on the technique known as **Gradient Pattern Analysis** (GPA) [1]. The technique consists of analyzing digital images (snapshots of a video, for example) based on the bilateral symmetry breaking of the underlying matrix gradient field. From the gradient field, metrics are extracted based on the geometry of the gradient field (via Delaunay triangulation) and also on the norm and phase matrices of the correspondent gradient field. GPA is therefore a set of computational operations on the matrix and on the gradient pattern of a digital image. In the last five years, the technique has been improved for automatic morphological classification associated with supervised training of machine learning models, especially with hybrid Deep Learning (CNN combined with RNN). In the scope of computer vision, GPA "sees" different morphologies in the gradient field and triangulation domains from where a set of metrics are extracted (see the example below of two galaxy images (spiral and elliptical) interpreted by GPA. Hands-on in Python will bring effective applications of GPA in some areas of interest such as image classification in astrophysics, solid-state physics and medicine.

1. INTRODUCTION

GPA allows quantifying patterns in the gradient field produced from digital image matrices. Metrics can be drawn from the triangulation between the vectors and also from the respective norms and phases of the gradient pattern. GPA works as a computational operator that is first applied to a matrix to obtain its respective gradient lattice (which can be total, symmetric or asymmetric). Figure 1 shows the GPA in the operation that allows obtaining the gradient moments from the asymmetric gradient lattice of a given matrix (or digital image). One successful example among many is the morphological classification of galaxies, which resulted in one of the most complete recent catalogs [2,3] (see Figure 2a).

The tutorial can be followed by advanced undergraduate students and beyond, including teachers and researchers interested in the topic. Due to the practical content of the course, we have been successful in applying it to a maximum of up to 20 participants. In the last edition, held at the University of Chile in 2018, the course had 45 participants. Many participants did not have a laptop and therefore compromised the assimilation of practical content, which is essential. However, the tutorial can be adapted as 2 hours short-course, with half for theory and half for practical demonstrations done collectively from two computers connected to the Google Colaboratory notebook. In this case, the audience can be much larger (thus depending more on the comfort and size of the class-room).

The GPA as an analytical method explores a new paradigm proposing to analyze, instead of the absolute values of the matrix amplitudes, the symmetry breaking in the gradient field of a given matrix. The symmetry to be broken, considered in the original version of the technique (Rosa, Sharma, Valdivia, 1999; Rosa et al., 2018), are the bilateral symmetries in relation to the vertical and horizontal axes. In the first operation of GPA, the Matrix Gradient Lattice is computed in the x-direction and the y-direction, whenever possible, with Central Finite Difference. The boundaries are measured with forward and backward finite differences. The gradient lattice (GL) of a matrix is composed of V vectors of which a portion V_A is asymmetric (bilaterally symmetric pairs in the gradient lattice are those that, for a given tolerance, have the same module and opposite phases). After obtaining the GL from a matrix, the first three gradient moments are calculated as follows:

$$G_1 = \frac{T_A - V_A}{V_A} \quad (1)$$

where T_A is the number of edges that represent the connection between the V_A asymmetric vectors obtained from the Delaunay triangulation performed on the GL. The measure, G_2 , based on the norms of each asymmetric vector v_i , is defined as:

$$G_2 = \frac{V_A}{V} \left(1 - \frac{|\sum_{i=0}^{V_A} v_i|}{2 \sum_{i=0}^{V_A} |v_i|} \right) \quad (2)$$

Note that if all vectors are symmetrical, $G_2 = 0.0$. The measure, G_3 , based on the phase of each asymmetric vector v_i , depends on the vector phases (ϕ) and the angle of each vector with respect to the symmetry center (θ) of the matrix, and is defined as:

$$G_3 = \frac{1}{2} \left(\frac{V_A}{V} + \frac{1}{2V_A} \sum_{i=0}^{V_A} (u_i \cdot v_i + 1) \right) \quad (3)$$

Note that, the dot product is maximum ($u \cdot v = 1$) when the position of the vector is aligned with its phase, zero when they are perpendicular, and minimum ($u \cdot v = -1$) when they are opposite. One of the possible formulations for G_4 , less usual than the other gradient moments, can be found in [5]. A formula that is often useful is the information entropy's formula that associates the values of norms and phases:

$$G_4 = \sum_i P(v_i) \cdot \log\left(\frac{1}{P(\phi_i)}\right) \quad (4)$$

Then G_1 is based on the entire Gradient Lattice, G_2 on the norms, G_3 on the phases, and finally G_4 on the complex composition using G_2 and G_3 . That is those measures are called Gradient Moments.

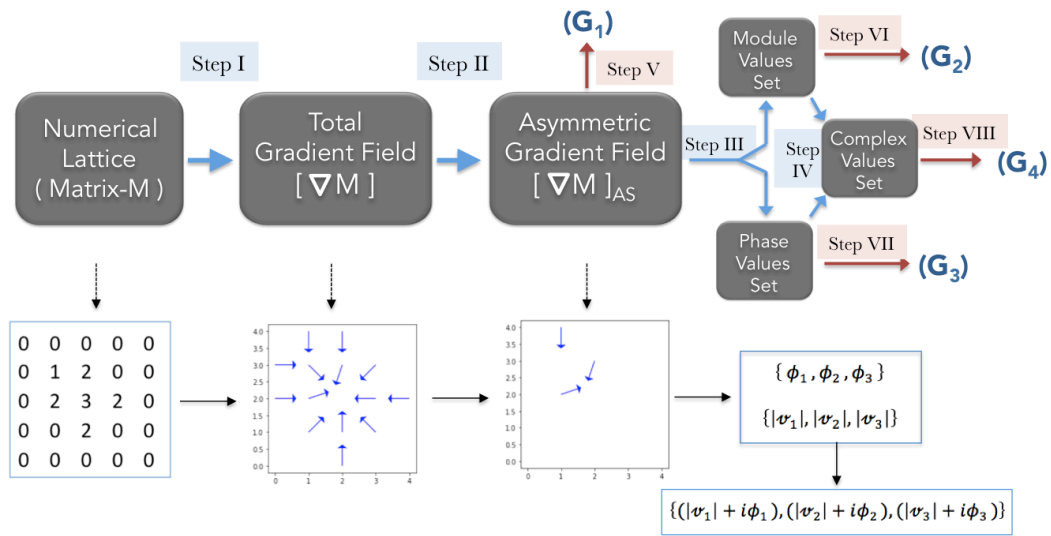


Figure 1 – The GPA as a computational operator in order to obtain metrics (the gradient moments) from a given numerical lattice as input.

2. PLANNED MATERIAL TO ATTENDEES

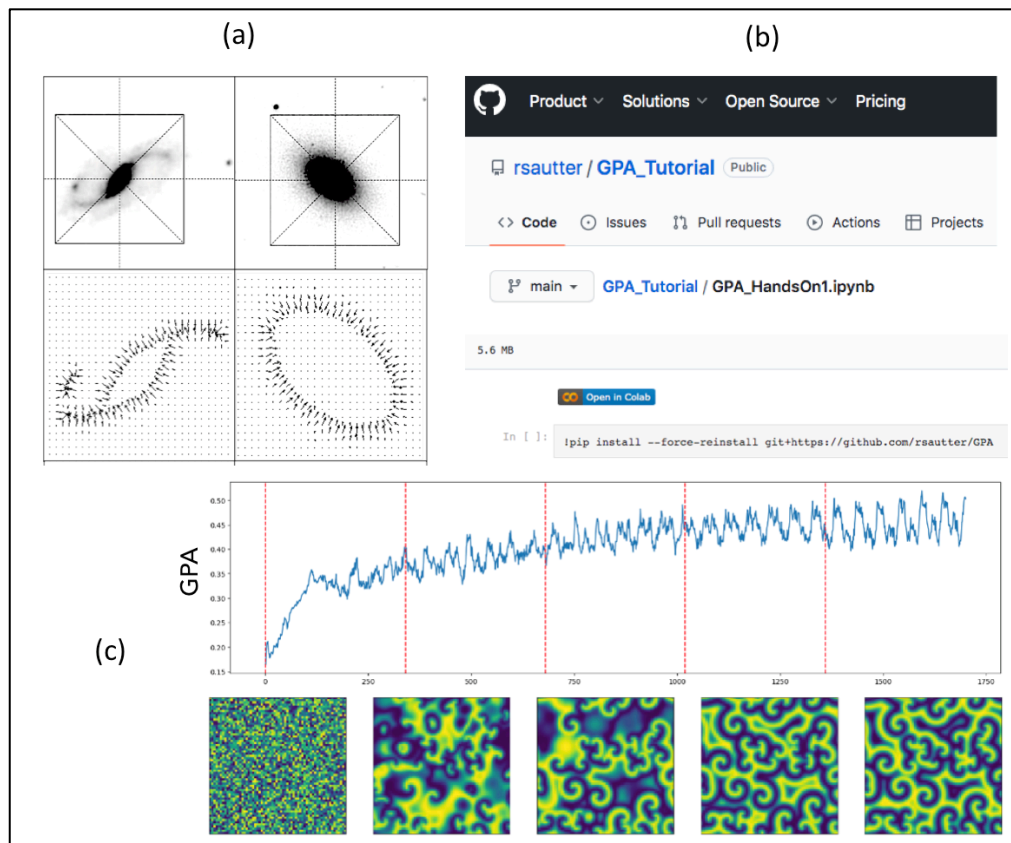


Figure 2: (a) GPA on Galaxy Classification [From 1,2]. (b) The Github Python Content. (c) A Dashboard example for monitoring pattern formation from GPA.

Participants will receive material deposited on Github (Fig. 2b), which includes a GPA notebook in Python. For this edition we will prepare a 40-page handout (which is part of a book on the subject to be released soon by Springer).

3. TOPICS TO BE COVERED

1- THE BILATERAL ASYMMETRY GPA OPERATION: THE ROLE OF GRADIENT PATTERN AND TRIANGULATION IN VISUAL ANALYTICS AND COMPUTER VISION [2,3].

2- DESCRIBING GPA ALGORITHM IN GOOGLE COLAB: A BRIDGE BETWEEN IMAGE PROCESSING AND COMPUTER VISION.

3- HANDS-ON APPLICATION EXAMPLES [3-16].

4- CONCLUDING REMARKS ON GPA AND CHALLENGES ON KAGGLE.

ACKNOWLEDGEMENT: The authors thank the support from the Brazilian Government Institutions as Lab for Computing and Applied Mathematics – National Institute for Space Research (INPE) and CNPq.

Relevant publications on the tutorial's subject:

[1] https://en.wikipedia.org/wiki/Gradient_pattern_analysis

[2] Barchi, P.H.; de Carvalho, R.R.; Rosa, R.R.; SauVer, R.A.; Soares-Santos, M.; Marques, B.A.D.; Clua, E.; Gonçalves, T.S.; de Sá-Freitas, C.; Moura, T.C. Machine and Deep Learning applied to galaxy morphology - A comparative study Astronomy and Computing, 2020 DOI: [10.1016/j.ascom.2019.100334](https://doi.org/10.1016/j.ascom.2019.100334)

[3] R. R. Rosa; R R de Carvalho; R A SauVer; P H Barchi; D H Stalder; T C Moura; S B Rembold; D R F Morell; N C Ferreira Gradient pattern analysis applied to galaxy morphology, 2018 Monthly Notes of the Royal Astronomical Society: Letters, DOI: [10.1093/mnrasl/sly054](https://doi.org/10.1093/mnrasl/sly054)

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